BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021 • Paper Code: DIK-HSSC-XII-2021 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 20 | **Paper Code:** DIK-HSSC-XII-2021 (A)

Q1. Choose the correct option i.e. A,B,C,D and fill it up in the correct bubble in the given MCQs sheet.

Qi. Which one of the following term is not related to term ketohexose?

(A) Aldehyde
(B) Ketone
(C) Six
(D) Sugar

Qii. Ethene and Ethyne can be distinguished by _____

(A) Anh- − $Alcl_3$
(B) Br_2 in CCl_4
(C) Cold H_2SO_4
(D) $AgNO_3$ in NH_4OH

Qiii. Ketone are les reactive than aldehyde towards nucleophilic attack due to _____

(A) Steric effect
(B) Electronic effect
(C) Resonance effect
(D) Both A and B

Qiv. Which of the following substances is not involved in photochemical smog formation?

(A) NO
(B) OI
(C) SO_2
(D) None

Qv. Salt _____ do not impart any colour to the flame.

(A) Na
(B) Li
(C) Be
(D) Ba

Qvi. The major component of bronze is _____

(A) Cu
(B) Zn
(C) Sn
(D) Fe

Qvii. The oxidation state of cobalt in is _____

(A) 1
(B) Zero
(C) 2
(D) 3

Qviii. All of the following are organic except_____

(A) Metal cyanides
(B) Urea
(C) Methane
(D) Acetic acid

Qix. Thiophene belongs to which class of organic compound?

(A) Heterocyclic
(B) Acyclic
(C) Alicyclic
(D) Homocyclic

Qx. Alcohols show no chemical reaction with _____

(A) Water
(B) Aqueous
(C) Sodium metal
(D) Both A and B

Qxi. Covalent character _____ down the group in tetra halides of 4th A elements.

(A) Increases
(B) Decreases
(C) Remain unchanged
(D) First decreases then increases

Qxii. Which of the following is functional group of acid halides?

(A) $-COOR$
(B) $-NH_2$
(C) $-COX$
(D) $-COOH$

Qxiii. Carboxylic group is so named because it contain a carbonal group and _____ group.

(A) Aldehyde
(B) Ketone
(C) Hydroxyl
(D) Carbon dioxide

Qxiv. An electron pair donor is classified as _____

(A) Lewis acid
(B) Lewis base
(C) Bronsted acid
(D) Arrhenius acid

Qxv. _____ test is used to identify aldehydes.

(A) Flame
(B) Baeyer's
(C) Lucas
(D) Tollen's

Qxvi. The hydrolysis of an ester in presence of alkali (NaOH) is known as _____

(A) Saponification
(B) Decarboxylation
(C) Esterification
(D) Trans esterification

Qxvii. Carotenes belong to family of _____

(A) Carbohydrate
(B) Protein
(C) Lipids
(D) Enzymes

Qxviii. Alpha helix and beta-pleated sheets are _____ structure of proteins.

(A) Primary
(B) Secondary
(C) Tertiary
(D) Quaternay

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 40

Q2. Attempt any Ten questions of the following. All questions carry equal marks

i. What is the importance of protein in our daily life? Give at least four points.

ii. Acetic acid is sometimes known as glacial acetic acid. Why?

iii. How hydrazine is prepared from aldehyde?

iv. How ethers can be synthesized from alcohol?

v. Define homologous series and give its characteristics.

vi. Give the reaction of Grignard reagent with: (i) Formaldehyde (ii) Carbon dioxide

vii. What is functional group? Give example.

viii. Describe the main products obtained by destructive distillation of coal.

ix. Write a short note on variable oxidation state of transition elements.

x. How beryllium differs from the member of its group?

xi. Draw the geometry of coordination number 4, 5 and 6 complexes.

xii. Write down the main function of ultraviolet and visible (UV/VIS) spectroscopy.

xiii. Why modern methods of analysis are superior to classical methods of analysis?

SECTION – C

Maximum Marks: 27

Note: Note: Attempt any three Question. Each question carries equal marks.

Q3 (a) Define inert pair effect. Why Sn and Pb use it?

(b) Discuss the colour of complex compounds with example.

Q.4 (a) What are alkene? How they are prepared from alcohols and alkylhalides?

(b) How benzene react with the following: i) CH_3Cl ii) $CH_3 - CO - Cl$ iii) HNO_3 iv) H_2SO_4

Q 5 (a) Define enzymes. Discuss the role of enzyme in digestion of protein fats and carbohydrates.

(b) Discuss the reaction of group IIA elements with oxygen.

Q. 6 (a) Compare the acidic behaviour of phenol with aliphatic alcohols.

(b) Write a note on any two of the following: (i) Acidity of terminal alkyne (ii) Photochemical smog (ii) Clemenson and Walf-Kishner's reduction reaction.